

DBA-MA L3 — IAEA Trigger Framework

Trigger Framework for Military Action Design Basis Accident

*Observable Indicators and Decision Criteria for Transition to
Autonomous Safe Shutdown Condition*

Version 3.0

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A Child Document Under the DBA-MA International Framework

1. Purpose and Scope

1.1 Purpose

This document establishes the Trigger Framework for the Military Action Design Basis Accident (DBA-MA). It defines the observable indicators, tiered threat levels, and decision criteria that govern the transition of a nuclear power facility from normal operations through preparation activities to controlled shutdown and achievement of the Autonomous Safe Shutdown Condition (ASSC). The Trigger Framework is a companion document to the DBA-MA Accident Description (DBA-MA L2) and is incorporated by reference into the DBA-MA operational requirements.

This document is an international reference document within the DBA-MA series. It applies to all national implementations that adopt the DBA-MA framework. National child documents adapt the authority structures, notification pathways, and jurisdictional references of this framework to each country's regulatory environment. The analytical content—observable indicator categories, tier structure, and decision logic—is jurisdiction-neutral and applies across all national implementations.

Implementation guidance for establishing the monitoring sources, information pathways, and data-sharing agreements necessary to observe and act on the indicators defined in this framework is provided in the companion Monitoring Sources and Data-Sharing Guidance (DBA-MA L4).

1.2 Operational Context

The Trigger Framework addresses a decision problem unique to the military action DBA: the facility operator must determine when to initiate controlled shutdown of a generating asset that may be critically needed for national defense and civilian welfare, based on indicators that an attack may be imminent. This decision involves competing obligations—the obligation to maintain electrical generation and the obligation to protect the public from radiological consequences of an uncontrolled event—and must be made under conditions of uncertainty.

The framework resolves this dilemma by establishing pre-defined, observable criteria that, when met, authorize and require specific actions. This approach removes the burden of judgment under pressure from individual operators and plant management, replacing it with a proceduralized response to defined conditions. This is directly analogous to the trigger criteria used by nuclear facilities in hazard-prone areas, where the decision to shut down is based on defined threat thresholds rather than on the individual operator's assessment of conditions.

1.3 Classification Considerations

The general structure of the Trigger Framework, including the tier definitions, action categories, and decision authority framework, is suitable for public release. Specific indicator thresholds—

including defined proximity criteria for adversary force movements, specific intelligence indicators, and classified threat characterizations—may require handling under applicable national security classification law. Classified parameters will be documented in classified annexes to this framework, maintained under the applicable national security framework.

2. Tiered Threat Structure

The Trigger Framework is organized into three tiers of escalating threat, each associated with specific observable indicators and required facility actions. The tiers are cumulative: actions required at a lower tier carry over to higher tiers unless superseded. The tiered structure parallels the watch/warning model used in existing nuclear industry preparedness frameworks.

2.1 Tier 1: Strategic Warning

2.1.1 Analogy

Tier 1 is analogous to the onset of a hazard season for a facility in a hazard-prone area, or to the issuance of a distant watch for an approaching storm. Conditions indicate an elevated probability of military action, but no imminent threat to the facility has been identified.

2.1.2 Observable Indicators

Tier 1 is activated when one or more of the following conditions are observed or communicated to the facility through established channels.

Increased nation-state cyber reconnaissance against energy sector targets, including the facility's own information technology and operational technology networks, or against energy sector targets in the facility's region or nation.

A significant and sudden increase or decrease in reporting of state-sponsored offensive cyber operations worldwide. Both an increase (indicating heightened activity) and a decrease (indicating potential operational security measures preceding an attack) are relevant indicators.

Escalating political or military tensions between the host nation and a state-level adversary possessing conventional military capability sufficient to conduct operations against nuclear facilities.

National defense condition changes that reflect an elevated threat posture, including changes to national alert status or equivalent defense readiness indicators recognized in the host nation's national security framework.

National security authority assessments, communicated through established channels, indicate a heightened likelihood of military action against the host nation or its critical infrastructure.

Intelligence reporting of adversary acquisition or forward deployment of attack drones in quantities consistent with a campaign against infrastructure, or evidence of adversary investment in AI-enabled drone capabilities.

2.1.3 Required Actions

Upon activation of Tier 1, the facility shall execute the following preparation activities while continuing normal power generation operations.

Verify that all consumable inventories (diesel fuel, lubricating oil, water treatment chemicals, and other materials identified in DBA-MA L2) are at or above levels specified for DBA-MA readiness. Initiate procurement to replenish any shortfalls.

Verify readiness of all systems designated for ASSC operations, including systems that may not be routinely operated. Conduct functional tests as specified in DBA-MA maintenance procedures.

Review and brief operating staff on military action response procedures, including Phase 2 transition procedures and ASSC maintenance procedures.

Establish or verify communication channels with intelligence, military command, and emergency management authorities as specified in facility-specific implementing procedures and in the applicable national child document.

Review and confirm pre-positioned emergency equipment and materials. Stage additional materials as specified in DBA-MA preparation procedures.

Notify the applicable national regulatory authority of Tier 1 activation through established reporting channels, as specified in the national child document.

2.2 Tier 2: Elevated Threat

2.2.1 Analogy

Tier 2 is analogous to a watch being upgraded to a warning, with the projected event track indicating probable impact on the facility's area. Military action against the nation or region is assessed as probable within a defined timeframe.

2.2.2 Observable Indicators

Tier 2 is activated when one or more of the following conditions are observed or communicated, in addition to or superseding Tier 1 indicators.

Declared hostilities against the host nation or its treaty allies by a state-level adversary, including formal declarations of war, initiation of hostilities without formal declaration, or military actions constituting de facto hostilities.

Military strikes against the host nation or allied nations, even if they do not directly target the facility or its region. Strikes against any territory of the host nation or its allies indicate a level of conflict that places all critical infrastructure at risk.

Nation-state cyberattacks against specific commercial or infrastructure targets within the host nation, particularly those targeting energy sector operations, grid management systems, or industrial control systems.

Nation-state kinetic attacks against specific commercial or infrastructure targets within the host nation, including attacks on energy generation, transmission, or distribution infrastructure.

An imminent nation-state cyber or physical activity that places core critical functions across energy or other critical infrastructure sectors at serious risk of degradation or disruption, as assessed by intelligence or national security authorities.

Confirmed adversary force movements within a defined proximity to the facility. The specific distance threshold is classified and documented in the classified annex to this framework, maintained under the applicable national security framework.

Confirmed strikes by uncrewed aerial systems against energy infrastructure within the host nation, reconnaissance by uncrewed aerial systems over or near the facility, or evidence of adversary staging of such systems within range of the facility.

2.2.3 Required Actions

Upon activation of Tier 2, the facility shall execute the following actions while maintaining power generation operations unless and until Tier 3 criteria are met.

Complete all Tier 1 actions that have not yet been accomplished. Top off all consumable inventories to maximum capacity.

Recall and stage additional operating personnel as specified in the facility's DBA-MA staffing plan. This may include recalling off-duty staff, activating emergency staffing agreements, and pre-positioning personnel needed for the Phase 2 transition and Phase 3 ASSC maintenance.

Begin power reduction and controlled cooldown if operationally appropriate and consistent with grid stability obligations. The decision to begin early power reduction at Tier 2 (before Tier 3 activation) rests with plant management in consultation with the national regulatory authority and grid operator. It should be based on an assessment of the time available before Tier 3 conditions develop.

Activate coordination with off-site emergency response organizations while communications and access remain available. Conduct final coordination with local emergency management on protective action recommendations and public notification procedures.

Secure or protect all documentation, records, and electronic data necessary for Phase 4 recovery assessment. This includes creating secure backups of safety analysis documentation, operating procedures, equipment records, and licensing basis documents.

Notify the applicable national regulatory authority and appropriate national authorities of Tier 2 activation, as specified in the national child document. Establish continuous communication with the national regulatory authority's operations center.

2.3 Tier 3: Imminent Threat

2.3.1 Analogy

Tier 3 is analogous to a warning with projected impact at the facility site, under which shutdown is mandatory. Military action against the facility or its immediate region is assessed as imminent or is underway.

2.3.2 Observable Indicators

Tier 3 is activated when one or more of the following conditions are observed, in addition to or superseding Tier 2 indicators.

Nation-state kinetic attacks against infrastructure targets in the facility's region, indicating that military operations have reached the facility's geographic area.

Adversary forces confirmed within a defined proximity to the facility. The specific distance threshold is classified and documented in the classified annex.

Confirmed targeting or disruption of the electrical transmission infrastructure serving the facility, indicating deliberate action to isolate the facility from the grid.

Direct military engagement (ground, air, or missile) in the facility's vicinity, where "vicinity" is defined by classified distance criteria in the classified annex.

National security authority assessment, communicated through established channels, of imminent attack on the facility or on critical infrastructure in the facility's region.

Loss of off-site power concurrent with any Tier 2 indicator, where the loss of power is assessed as potentially attributable to hostile action rather than routine grid events.

Strikes by uncrewed aerial systems against infrastructure in the facility's region, detection of surveillance drones directly over the facility, or drone attacks on the facility's off-site power infrastructure. Because a drone campaign can develop within hours, these indicators are the principal reason the standing authority in Section 3.1 must not depend on external authorization.

2.3.3 Required Actions

Upon activation of Tier 3, the facility shall execute a transition to the Autonomous Safe Shutdown Condition. This action is mandatory and is not subject to deferral for grid stability or generating capacity considerations. The nuclear safety requirement to prevent uncontrolled radiological release from a facility that is operating at the time of a military attack takes precedence over the generating capacity requirement.

Initiate emergency shutdown (reactor trip) if the facility has not already begun a controlled shutdown during Tier 2. Execute Phase 2 transition procedures to achieve the ASSC as rapidly as plant conditions permit.

Transition to minimal-staffing operational posture as specified in DBA-MA procedures. Non-essential personnel should be released from the site to the extent practical and safe.

Activate all autonomous safety systems designated for ASSC operations.

Transmit the final status report to the national regulatory authority and designated national authorities. Establish communications on backup channels designated for use in degraded conditions.

Execute all final site preparation activities specified in DBA-MA procedures, including securing access points, activating physical protection measures for safety systems, and completing any remaining material staging.

3. Decision Authority

3.1 Authority for Tier Activation

The authority to activate each tier of the Trigger Framework and execute the associated required actions must be clearly assigned and pre-authorized within each national implementation. The DBA-MA Trigger Framework must not create a situation in which, during a crisis, the operator must seek authorization to act from authorities that may be unreachable. National child documents specify the authority structure applicable within each jurisdiction. The general principles governing the allocation of authority are as follows.

Tier 1 activation may be initiated by plant management based on independently observable indicators, or by notification from the national regulatory authority, national security authority, or national defense authority. Plant management has standing authority to activate Tier 1 based on any observable indicator without prior approval from the external authority.

Tier 2 activation may be initiated by plant management based on observable indicators or by direction from the national regulatory authority in coordination with national security and defense authorities. Plant management has standing authority to activate Tier 2 without prior regulatory approval when observable indicators are met, though prompt notification to the regulatory authority is required.

Tier 3 activation and the mandatory shutdown decision may be initiated by plant management or directed by the national regulatory authority. Critically, plant management must have pre-authorized standing authority to initiate Tier 3—including emergency shutdown—based on observable indicators, without waiting for regulatory direction. The rationale is that conditions requiring action may develop faster than external communication can allow, and the on-site operator is best positioned to observe local conditions.

National authority and plant authority combine in one direction only. The plant's standing authority to activate any tier, including Tier 3 and the transition to the ASSC, on its own observable indicators is a floor: the plant may always move toward a safer state without external authorization. National authority may direct a tier change, but its direction is never a precondition for the plant to protect itself. This is the principle of local sovereignty, set out in the parent framework, Sections 5.1 and 7.4, applied to decisions. Where any part of the decision chain is supported by automated or AI-enabled systems, the governance of those systems, including which actions they may take and which require human authorization and by whom, is addressed in CB-Annex-Nuclear.

3.2 Information Access Requirements

Several trigger indicators depend on information that may not currently be available to nuclear plant operators through established channels. These include classified national security authority assessments of adversary intent and capability; real-time military situational awareness, including adversary force positions; specific threat information from national security agencies; and national defense condition changes that may not be publicly announced in real time.

Implementation of the Trigger Framework requires establishing information-sharing agreements and communication channels between the facility and the relevant national authorities. These agreements must specify what information will be shared, through what channels, with what classification protections, and with what timeliness. The information channel must be survivable under the threat conditions it is designed to detect—it cannot depend on communications infrastructure that may itself be targeted.

Detailed guidance on the specific monitoring sources, government programs, and data-sharing agreements required to support each observable indicator is provided in the Monitoring Sources and Data-Sharing Guidance (DBA-MA L4). That document includes an agreement-readiness self-assessment that each facility should complete to identify gaps in its current information-access posture. National child documents adapt this guidance to the specific information-sharing architecture of each jurisdiction.

3.3 Coordination with Grid Operations

The Trigger Framework acknowledges the tension between nuclear safety and grid reliability during periods of national crisis. At Tier 1 and Tier 2, the facility continues to generate while preparing for a potential shutdown. The decision to begin early power reduction at Tier 2 is made in coordination with the grid operator and the national regulatory authority, balancing grid needs against preparation time requirements.

At Tier 3, the nuclear safety requirement is paramount. The rationale for this precedence is explicit: an uncontrolled radiological release resulting from a military attack on an operating reactor creates consequences—long-term land contamination, public evacuation, and radiological

exposure—that far exceed those of the loss of generating capacity from a controlled, preemptive shutdown. Grid operators should be included in the Tier 2 notification process to prepare for the potential loss of the facility's generating capacity. Still, grid reliability considerations cannot override the Tier 3 shutdown requirement.

4. Standing Preparedness Requirements

The following requirements apply at all times, regardless of the current threat tier, and are analogous to the continuous monitoring performed by facilities in hazard-prone areas. Implementation details, including the specific organizations and programs applicable in each jurisdiction, are addressed in DBA-MA L4 and in the applicable national child document.

Each facility shall participate in threat information sharing through the appropriate national or regional electricity sector information-sharing organization to maintain awareness of cyber and physical threat developments relevant to the energy sector.

Each facility shall maintain relationships and communication channels with the military, intelligence, and emergency management organizations identified in the facility-specific Trigger Framework implementation procedures. These relationships shall be exercised through periodic drills and coordination meetings.

Each facility shall conduct exercises of the DBA-MA response procedures, including Trigger Framework activation, Phase 2 transition, and ASSC maintenance, at intervals to be determined by the applicable national regulatory authority. Exercises should include participation by relevant external agencies to the extent practicable.

Each facility shall maintain the consumable inventories, equipment readiness, and staffing plans specified in DBA-MA L2 at all times, as a baseline preparedness posture independent of the current threat assessment.

Each facility shall maintain a capability to detect and characterize uncrewed aerial systems as a standing element of its monitoring posture, since the drone indicators at each tier cannot be observed without it. A multi-sensor approach integrating radar, acoustic, radio-frequency and electro-optical sensors is the most robust; as AI-enabled drones increasingly operate without detectable control-link emissions, radar and electro-optical detection become the primary modalities. DBA-UxS, Section 4.3, and DBA-MA-UxS-NA set out the detection considerations.

Uncrewed surface and undersea vehicles. Facilities with waterfront exposure or maritime access routes shall also monitor for uncrewed surface vessel and uncrewed undersea vehicle activity. Until indicators for those systems are incorporated into the tier structure in a future revision, the threat characterization and detection provisions for them in DBA-UxS, Sections 5 and 6, and in DBA-MA-UxS-NA shall be applied as supplementary guidance alongside the tiers defined in this document.

5. References

5.1 DBA-MA Series Documents

DBA-MA International Framework (L1) — Expanding the Design Basis for Nuclear Power Plants to Address Military Action Against Nuclear Facilities: An International Framework. Eclectic Technologies.

DBA-MA L2 — Design Basis Accident: Military Action Against Nuclear Facilities. Formal Accident Description and Design Requirements. Eclectic Technologies.

DBA-MA L4 — Monitoring Sources and Data-Sharing Guidance for DBA-MA Trigger Observables. Eclectic Technologies.

DBA-UxS — Uncrewed Systems: Threat Considerations for Critical Infrastructure. Eclectic Technologies.

DBA-MA-UxS-NA — Uncrewed Systems: Nuclear Sector Implementation Annex. Eclectic Technologies.

CB-Annex-Nuclear — Command Broker: Nuclear Sector Annex. Application of the Cross-Sector Command Broker Requirement to Nuclear Facilities. Eclectic Technologies.

5.2 IAEA References

IAEA Safety Fundamentals No. SF-1 — Fundamental Safety Principles (2006).

IAEA Safety Standards Series No. SSR-2/1 — Safety of Nuclear Power Plants: Design (Rev. 1, 2016).

IAEA Nuclear Security Series No. 10-G (Rev. 1) — National Nuclear Security Threat Assessment, Design Basis Threats and Representative Threat Statements (2021).

IAEA, Nuclear Safety, Security and Safeguards in Ukraine — Director General summary reports and updates (2022–present).

5.3 National Implementation References

National regulatory and legislative references applicable to each implementation of this document are identified in the corresponding national child document. The United States national references are set out in DBA-MA L3, United States National Implementation.

[Note: Classified annexes containing specific threat characterization parameters, distance thresholds, intelligence indicators, and security-protected information will be maintained separately under appropriate access controls within each national jurisdiction.]

Revision History

Revision notes for versions earlier than 3.0 were not carried in the drafts and are not reconstructed here.

Revision Note (v3.0, September 2026): Issued as the first final version, aligned with the DBA-MA International Framework, Version 2.0. Removes the draft status and adds the citation block, stamped with the reserved version DOI 10.5281/zenodo.22945421. Adds the drone observables for each tier from the parent framework, Section 7.5, and a standing requirement for drone detection. States in Section 3.1 that the plant’s standing authority is a floor that national direction may add to but never precondition, with a pointer to CB-Annex-Nuclear. Moves the uncrewed surface and undersea vehicle provision, formerly an unnumbered “4.5” placed after the references, into Section 4 and corrects its reference from DBA-UxS Section 9, which governs review cycles, to Sections 5 and 6. Cites series instruments by designation without version numbers and names the Nuclear Security Series document relied on.